	Tens	Ones	Value
Given Number	x	y	$10x + y$
New Number (after reversal)	y	x	$10y+x$

Given, Original number – reversing number = 27

$$(10x + y) - (10y + x) = 27$$

$$10x - x + y - 10y = 27$$

$$9x - 9y = 27$$

$\Rightarrow$ $$x - y = 3 \quad \ldots (2)$$

Also from (1), $y = 5 - x$ ... (3)

Substitute (3) in (2) to get $x - (5 - x) = 3$

$$x - 5 + x = 3$$

$$2x = 8$$

$$x = 4$$

Substituting $x = 4$ in (3), we get $y = 5 - x = 5 - 4$

$$y = 1$$

Thus, $10x + y = 10 \times 4 + 1 = 40 + 1 = 41$.

Therefore, the given two-digit number is 41.

Verification :

sum of the digits = 5

$x + y = 5$

$4 + 1 = 5$

$5 = 5$ true

Original number – reversed number = 27

$41 - 14 = 27$

$27 = 27$ true

Exercise 3.11

1. Solve, using the method of substitution

(i) $2x - 3y = 7;\ 5x + y = 9$ (ii) $1.5x + 0.1y = 6.2;\ 3x - 0.4y = 11.2$

(iii) 10% *of* $x + 20\%$ *of* $y = 24;\ 3x - y = 20$ (iv) $\sqrt{2}\,x - \sqrt{3}\,y = 1;\ \sqrt{3}\,x - \sqrt{8}\,y = 0$

2. Raman's age is three times the sum of the ages of his two sons. After 5 years his age will be twice the sum of the ages of his two sons. Find the age of Raman.

3. The middle digit of a number between 100 and 1000 is zero and the sum of the other digit is 13. If the digits are reversed, the number so formed exceeds the original number by 495. Find the number.

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